

Managing Credentials in Jenkins

To securely manage credentials in Jenkins, follow these steps:

1. Storing Credentials:

- Navigate to **Manage Jenkins > Security > Manage Credentials**.
- Choose the appropriate credentials domain (e.g., **Global** for credentials used across all Jenkins jobs).
- Click **Add Credentials** to store new credentials, such as GitHub tokens, Docker registry credentials, or SonarQube API tokens.

2. Types of Credentials:

- **Username and Password:** Commonly used for services that require login (e.g., GitHub).
- **Secret Text:** Often used for API tokens (e.g., Docker or SonarQube tokens).
- **SSH Username with Private Key:** Useful for SSH-based authentication.

3. Using Credentials in Jobs:

- In your Jenkins job configuration, go to the **Source Code Management** section (for GitHub or Git) or other service configurations (Docker, SonarQube, etc.).
- Choose the credential type that matches your setup. For example, if linking Jenkins to a Git repository, select the stored GitHub credentials for cloning or pulling code.

Plugins in Jenkins

Plugins enhance Jenkins' functionality, adding new features based on your project needs. To manage plugins:

- Go to **Manage Jenkins > Manage Plugins**.
- Browse the available plugins or install them based on your needs (e.g., GitHub, Docker, SonarQube, etc.).
- After installation, configure these plugins in the **System Configuration** section.

Operating Jenkins

- **GUI:** Jenkins has a web-based interface to manage jobs, pipelines, and configurations.
- **CLI:** You can also operate Jenkins through its Command-Line Interface (CLI). To access CLI commands:
 - Go to the **Jenkins CLI** section in the web interface.
 - Use the appropriate Jenkins CLI commands for tasks like managing jobs, credentials, and nodes.

Jenkins System Configuration

To configure Jenkins system-wide settings, follow these steps:

1. System Home Directory:

- Jenkins' home directory is usually located at `/var/lib/jenkins`. This directory contains all Jenkins configuration files, job data, and plugins.

2. Build Executors:

- Executors are the agents that run jobs in Jenkins. If you have 5 executors and 10 jobs, only 5 jobs will be built simultaneously, and the remaining 5 will be placed in the build queue.
- Ensure that you configure executors based on your system's resources (e.g., 5-8 executors for moderate workloads).
- Executors can be distributed between the **Master** and **Slave** nodes, forming a distributed build system. The master node manages the build while the slave nodes execute the tasks.

3. Quiet Period:

- The quiet period is the delay between the triggering of a build and its actual start. This ensures that Jenkins waits for a set amount of time before beginning the build, allowing for any last-minute changes to the job.

4. SCM (Source Code Management) Checkout Retry Count:

- Configure the number of retries Jenkins should make when checking out code from a version control system (like Git) in case of failure. This is useful to handle intermittent issues with GitHub connections or network instability.

5. Email Notifications:

- You can configure Jenkins to send email notifications when builds start, succeed, or fail.
- Go to **Manage Jenkins > Configure System > Email Notification** and input your SMTP server details for email automation.